Part I

Estimate the total number of patterns from the partial truncated observations

1 Notation and setup

we have

1. $N = \sum_{k=1}^{K} n_k$
2. K: unique patterns observed, with counts $n = \{n_1, \vec{n}_2, ...n_k\}$
3. [latent] S= true total number of patterns, unknown, $S \geq K$
4. [latent] $S-K$ number of patterns that existed but were not observed (zero counts)

We have observed data $\vec{n} = \{n_{1,} n_{2,} ... n_K\}$, where K is the number of observed patters. n_k indicates the counts of sample that belong to the pattern k. By Bayes rule we have:

$$P(S|\{n_{1,} n_{2,} ... n_K\}) = P(\{n_{1,} n_{2,} ... n_K\}|S)P(S)$$

Question is how to get the likelihood $P(\{n_{1,} n_{2,} ... n_K\}|S)$? Here we also use $\vec{m} = \{m_{1,} m_{2,} ... m_S\}$ to denote the full latent count vector over all S patterns, where m_s can be 0 for unobserved patterns while n_k is always nonzero for observed ones.

2 Likelihood estimate

So this really is the probablility that we observe $S - K$ category to have zero counts (in $\sum_k n_k$ samples) and observed data $\vec{n}$ from K categories.

The simplest way of getting this likelihood is assumign the Multinomial distribution for the counts (here $\{m_1, m_2, ...m_S\}$) over all S patterns.

$$P(m_1, m_2, ...m_S|S) = \frac{N!}{m_1!m_2!...m_S!} \prod_{s=1}^{S} p_s^{m_S}$$

where p_k is the probability of observing the type k. This is just Multinomial distribution PMF.

Now, some of these value is zero and we dont actually observe them. And we dont even know which of the S are NOT observed. We need to multiply something like $\binom{K}{S}$ to account for this fact. this way we can assume we observed the first Kcategories and the rest $S-K$ was just zero observations

$$P(\{n_1,n_2,...n_K\},K|S) = \binom{S}{K}\frac{N!}{n_1!n_2!...n_K!(0!)^{S-K}}\prod_{k=1}^{K}p_k^{n_k}\prod_{j=K+1}^{S}p_j^0$$

2.1 Uniform model

For simplicity we can probably assume $p_s = \frac{1}{S}$, $\forall s = 1,...S$. Then above becomes

$$P(\{n_1,n_2,...n_K\},K|S) = \binom{S}{K}\frac{N!}{n_1!n_2!...n_K!}\left(\frac{1}{S}\right)^N$$

Expand the combination term

$$\binom{S}{K} = \frac{S!}{K!(S-K)!}.$$

And we ignore the term that is NOT dependent on S – so remove $K!$ and the fraction of $N!$. We get

$$P(\{n_1,n_2,...n_K\},K|S) \propto \frac{S!}{(S-K)!}\cdot\frac{1}{S^N}$$

So,,,,, we have the form for the likelihood. now putting this back to our Bayes rule:

$$P(S|\vec{n},K) \propto \frac{S!}{(S-K)!S^N}\cdot P(S)$$

If we believe the prior is uniform ($P(S) \propto 1$) then

$$P(S|\vec{n},K) \propto \binom{S}{K}\left(\frac{1}{S}\right)^N \propto \frac{S!}{(S-K)!S^N}$$

So we can get the most likely S as well as full posterior by computing this on log scale (to avoide factorial blowing up):

$$\log P(S|\vec{n},K) = \log(S+1) - \log(S-K+1) - N\log S + const$$

to get full posterior we can subtract the max (for numerical stability), exponentiate and normalize, to get the posterior. MAP estimate is the S that maximize this.

2.2 Dirichlet-Multinomial with full S

Now we set Dirichlet as the prior for the probability. Here we assume even the zero counts are observed (i.e.,m_s can be zero). To avoid the confusion here we use θ_s to denote the probabiliy in the Multinomial so we have:

$$\vec{\theta}|\alpha \sim \text{Dirichlet}(\alpha) = \frac{1}{B(\alpha)} \prod_s^S \theta_s^{\alpha_s - 1}$$

$$P(\vec{m}|\vec{\theta}, S) = \text{Multinomial}(N, \theta) = \frac{N!}{\prod_s^S m_s!} \prod_{s=1}^S \theta_s^{m_s}$$

$$P(\vec{m}|\alpha, S) = \int P(\vec{m}|\theta, S) P(\theta|\alpha, S) d\theta$$

Substitute the expressions and we get

$$P(\vec{m}|\alpha, S) = \frac{N!}{\prod_s m_s!} \cdot \frac{1}{B(\alpha)} \int \prod_s^S \theta^{m_s + \alpha_s - 1} d\theta$$

Note the integral part is equal to the normalization constant of Dirichlet with parameters $\alpha_s + m_s$.

$$\int \prod_s^S \theta^{m_s - \alpha_s - 1} d\theta = B(\alpha + m)$$

Therefore,

$$P(\vec{m}|\alpha, S) = \frac{N!}{\prod_s m_s!} \cdot \frac{B(\alpha + \vec{m})}{B(\alpha)}$$

The beta functions are given by

$$B(\alpha) = \frac{\prod_s^S \Gamma(\alpha_s)}{\Gamma(\sum_s \alpha_s)}, \; B(\alpha + m) = \frac{\prod_s^S \Gamma(\alpha_s + m_s)}{\Gamma(\sum_s \alpha_s + N)}$$

Then

$$P(\vec{m}|\alpha, S) = \frac{N!}{\prod_s m_s!} \frac{\Gamma(\sum_s \alpha_s)}{\Gamma(N + \sum_s \alpha_s)} \prod_s^S \frac{\Gamma(\alpha_s + m_s)}{\Gamma(\alpha_s)}$$

Gamma function $\Gamma(a) = (a-1)!$. So far we simply gave the Dirichlet-Multinomial likelihood. So what about when we observed K categories?

2.3 Dirichlet-Multinomial with partially observed patterns K

So we follow the similar idea here, breaking down to the first K categories and last $S-K$ categories, and include $\binom{S}{K}$ term. Lets also assume $\alpha = \alpha_1 = \alpha_2 = ... = \alpha_S$.

$$P(\vec{n}, K|\alpha, S) = \binom{S}{K} \frac{N!}{\prod_k n_k!} \frac{\Gamma(S\alpha_s)}{\Gamma(N + S\alpha_s)} \prod_{k=1}^{K} \frac{\Gamma(\alpha_k + n_k)}{\Gamma(\alpha_k)} \prod_{j=K+1}^{S} \frac{\Gamma(\alpha_j + m_j)}{\Gamma(\alpha_j)}$$

Since $m_j = 0$, the last term is 1, so

$$P(\vec{n}, K|\alpha, S) = \binom{S}{K} \frac{N!}{\prod_k n_k!} \frac{\Gamma(S\alpha)}{\Gamma(N + S\alpha)} \prod_{k=1}^{K} \frac{\Gamma(\alpha + n_k)}{\Gamma(\alpha)}$$

Take the term that is dependent on S,

$$P(S|\vec{n}, K, \alpha) \propto P(\vec{n}, K|S, \alpha) \propto \binom{S}{K} \frac{\Gamma(S\alpha)}{\Gamma(N + S\alpha)} \prod_{k=1}^{K} \frac{\Gamma(\alpha + n_k)}{\Gamma(\alpha)}$$

This formula would be nice to estimate (MLE) α from observed ($\vec{n}$). And then plug in $\hat{\alpha}$ to get likelihood of S.

$$\hat{\alpha} = \arg\max_{\alpha} P(\vec{m}|\alpha, S) = \arg\max_{\alpha} \left[\frac{\Gamma(\sum_s \alpha_s)}{\Gamma(N + \sum_s \alpha_s)} \prod_{s}^{S} \frac{\Gamma(\alpha_s + m_s)}{\Gamma(\alpha_s)} \right]$$

But we dont know S..!! So what we can do is use K to replace S above. However, $\hat{\alpha}$ estimated this way will be assuming that there are K patterns in total hence. Specifically if we only observe K (and ignore the zero counts patterns) then that will make α goes to more uniform side, so α is inflated (larger than the actual value).

One way to do it is estimate $\hat{\alpha}(S)$ for each S, and plug that to get $P(\vec{n}, K|S, \hat{\alpha}(S))$. But thats weird, it understates the uncertainty in α.

2.4 Joint inference on α and S

So the sensible thing to do here seems to be jointly inferring the two parameters α and S.

$$P(S, \alpha|\vec{n}, K) \propto P(\vec{n}, K|S, \alpha)P(S, \alpha)$$

From the results above, the RHS is

$$P(\vec{n}, K|\alpha, S) = \binom{S}{K} \frac{N!}{\prod_k n_k!} \frac{\Gamma(S\alpha)}{\Gamma(N + S\alpha)} \prod_{k=1}^{K} \frac{\Gamma(\alpha + n_k)}{\Gamma(\alpha)}$$

The actual posterior over S would be

$$P(S|\vec{n}, K) = \int P(S, \alpha|\vec{n}, K)\ d\alpha$$

3 Estimate the required total sample size

From above results, we have

$$P(S|\vec{n}, K) = \int P(S, \alpha|\vec{n}, K)\ d\alpha$$

Lets say now, we want to get more samples s.t. the minimum amount of samples per categories are at least T. In other words, we want to determine the smallest N' samples s.t. we would exepct to see at least Tsamples per pattern with high probability. We use $\tilde{m_s}$ to denote the count of samples in pattern s, when N' total samples are drawn ($\sum_s \tilde{m_s} = N'$).

$$P(\tilde{m_s} \geq T \text{ for all } s = 1, ...S|K, N', \vec{n}) \geq 1 - \delta$$

So the posterior predictive:

$$P(\tilde{m_s}|K, N', \vec{n}) = \sum_{S=K}^{\infty} \int P(\tilde{m_s}|S, \alpha, N')P(S, \alpha|K, \vec{n})d\alpha$$

We already know the second terms inside the integral from the previous sections. The first term inside the integral is Dirichlet-Multinomial PMF, which is also derived in the previous section. SO the probability of interest would be:

$$P(\tilde{m_s} \geq T, \forall s|K, N', \vec{n}) = \sum_{S=K}^{\infty} \int P(\tilde{m_s} \geq T, \forall s|S, \alpha, N')P(S, \alpha|K, \vec{n})d\alpha$$

The first term is

4 Numerical example (make it sounds similar to the LC case)

Assume this: we observed 100 neurons in total. 20 of them project to spinal cord, 30 of them project to the cerebellum, 25 of them project to thalamus (thalamus only) and the rest of 25 project to the frontal cortex. $K = 4$. What is the total number of projection pattern we actually have here?

Lets investigate two cases: large α which gives more uniform across the patterns, and small α which gives more sparse (high entropy) results.

4.1 Large α

lets say our observation is $\vec{n} = \{20, 30, 25, 25\}$.

4.2 Small α

Lets say our observation is $\vec{n} = \{5, 3, 7, 85\}$